Instituto Tecnológico de Costa Rica
Escuela de Ingeniería en Computadores
Modelación Hardware/Software con Orientación a Objetos • CE-5507

Project 3 - Attributes Document

Interactive product with hardware and software

Smart Pill Reminder Box

Object-Oriented pill reminder system with a software app and hardware interface

System Design • ESP32 • Pill Reminder
C • C++ • Arduino IDE • Web Design

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Table of Contents

1. Introduction

This document presents the assessment of the **Engineering Tools (ET)** attribute for the Smart Pill Reminder Box project, developed for the CE-5507 Hardware/Software Modeling with Object-Oriented Design course at the Instituto Tecnológico de Costa Rica. The attribute evaluates the ability to create, select, apply, adapt, and extend modern engineering and information technology techniques, resources, and tools, including the investigation and modeling of complex engineering problems with an understanding of their associated limitations.

The Smart Pill Reminder Box is an embedded IoT device that assists older adults with cognitive impairment in following their medication routines. Its development required a combination of hardware components, communication protocols, modeling techniques, software methodologies, and manufacturing processes. This breadth makes it a suitable case for evaluating how engineering tools were selected, applied, and adapted throughout the design.

The remainder of this document is organized into three sections, following the structure required by the attribute. Section II details the selection of the techniques, resources, tools, and methods used in the project, stating the function and selection rationale of each. Section III provides evidence of the application of those tools across the development process. Section IV describes the cases where existing tools or methods were created, adapted, or extended to meet the specific requirements of the project, indicating their function and evidence of use.

2. Selection of Techniques, Resources, Tools, and Methods

This section lists the techniques, resources, tools, and methods selected for the project, grouped by domain. For each one, its function within the system and the rationale behind its selection are indicated.

2.1. Hardware and Electronic Components

Tool / Component	**Function**	**Selection Rationale**
ESP32 microcontroller	Central control unit that runs the reminder logic, reads inputs, drives outputs, and handles communication.	Includes integrated WiFi, removing the need for an external communication module, and provides enough GPIO pins for all sensors and actuators.
Magnetic reed switches	Detect whether each drawer is open or closed.	Allow the sensor to be concealed inside the structure, avoiding visible buttons that could confuse the target user.
Drawer LEDs	Indicate the correct drawer during a reminder.	Provide a localized, low-cost visual cue that improves accessibility for users with cognitive impairment.
Buzzer	Generate the audible alert during a reminder.	Adds an auditory channel that reinforces the reminder for users with attention or perception difficulties.
LCD screen	Display the medication name, drawer number, and confirmation or warning messages.	Delivers direct textual instructions, reducing ambiguity during the medication process.
Current-limiting and pull resistors	Protect the LEDs and stabilize the digital signal from the reed switches.	Required for safe and reliable operation of the output and input circuits.

Table 1: Selected hardware components and their selection rationale

2.2. Software, Communication, and Modeling

Tool / Method	Function	Selection Rationale
Object-oriented design	Structure the firmware and mobile application into classes with separated responsibilities.	Required by the course and well suited to mapping physical components and domain entities into a maintainable, extensible architecture.
MQTT protocol over WiFi	Carry configuration data and interaction feedback between the mobile application and the ESP32.	Lightweight publish-subscribe standard widely used in IoT healthcare systems, suitable for a home-use device.
Mobile application	Allow the caregiver to register medications, assign drawers, set schedules, and receive notifications.	The primary user may not be able to configure the device, so configuration is delegated to a caregiver-facing tool.
UML use case diagram	Model the interactions between the caregiver, the patient, and the system.	Clarifies actor responsibilities and system requirements early in the design.
UML class diagram	Describe the object-oriented structure of the application and the embedded software.	Documents the four-layer architecture and the relationships between classes.
State diagram (FSM)	Model the internal behavior of the ESP32 firmware across the reminder lifecycle.	Captures each operational mode and the events that trigger transitions, guiding firmware implementation.
LaTeX	Produce structured, reproducible technical documentation.	Provides precise control over tables, diagrams, and references for engineering reports.

Table 2: Selected software, communication, and modeling tools and their selection rationale

2.3. Manufacturing and Materials

Tool / Material	Function	Selection Rationale
3D modeling (CAD)	Define the geometry of the box and drawers before fabrication.	Allows dimensions and component placement to be validated before cutting physical parts.
Laser cutting	Manufacture the drawers and structural parts from sheet material.	Produces clean edges and accurate dimensions, enabling rapid prototyping that matches the technical drawings.
Transparent acrylic	Build the drawer compartments.	Diffuses LED light evenly, improving the visual feedback that identifies the selected drawer.
Plywood	Build the external structure and non-electrical parts.	Opaque to conceal wiring, with warm tactile and acoustic qualities suited to older adult users.

Table 3: Selected manufacturing tools and materials with their selection rationale

3. Application of Techniques, Resources, Tools, and Methods

This section provides evidence of how each selected tool was used during the development of the project.

Tool / Method	**Evidence of Use**
ESP32 microcontroller	Defined as the central node in the system architecture, connected to the reed switches as digital inputs and to the LEDs, buzzer, and LCD as outputs, and modeled in the firmware through the `DeviceController` class.
Magnetic reed switches	Distributed one per drawer in the component layout, modeled as the `ReedSwitch` class with `readState()` and `onStateChange()`, and used to confirm drawer interaction in the state diagram.
Drawer LEDs, buzzer, LCD	Modeled as the `LED`, `Buzzer`, and `LCDScreen` classes in the hardware layer and triggered together in the "Reminder Active" state to provide multimodal guidance.
Object-oriented design	Applied throughout the class diagram, which organizes the system into mobile application, domain, device services, communication, and hardware layers.
MQTT over WiFi	Implemented through the `MQTTClient`, `ConfigMessage`, and `NotificationEvent` classes, used to send configuration from the app and feedback from the device.
Mobile application	Modeled through the `MobileApp`, `Caregiver`, and `AlertNotification` classes and shown as the configuration and monitoring endpoint in the operation flow and architecture diagrams.
UML use case diagram	Produced for the project, defining the caregiver and patient actors and their respective interactions with the application and the device.
UML class diagram	Produced as the full object-oriented model of the system, including attributes, methods, and inter-layer relationships.
State diagram (FSM)	Produced to model the firmware lifecycle through the Idle, Reminder Active, Drawer Open, Timeout, Interaction Complete, and Sending Feedback states.
LaTeX	Used to author the design document, including the materials table, the embedded state diagram, and all cross-referenced figures.
3D modeling and laser cutting	Used to model and fabricate the demonstration unit; the drawers and structure are laser cut from the modeled geometry.
Acrylic and plywood	Assigned to the drawers and external structure respectively, as documented in the materials table of the design document.

Table 4: Evidence of application of the selected tools and methods

4. Creation or Adaptation of Techniques, Resources, Tools, and Methods

This section describes the cases where standard tools or methods were created, adapted, or extended to address the specific requirements of the project. For each one, its function and evidence of use are indicated.

Creation / Adaptation	**Function**	**Evidence of Use**
Reed switch repurposed as a drawer-interaction sensor	Detect medication-compartment interaction instead of its conventional door or security use.	Each drawer pairs a fixed reed switch with a moving magnet; the open-then-close sequence is used in the firmware as a proxy for medication interaction.
Drawer-interaction confirmation method	Confirm that the user acted on the correct compartment without invasive sensing.	The state diagram requires the selected drawer to be opened and then closed again before the dose is logged as taken.
Custom bidirectional MQTT message layer	Adapt the generic publish-subscribe pattern into a domain-specific configuration and feedback exchange.	The `ConfigMessage` (drawer, medication, schedule) and `NotificationEvent` (event type, drawer, timestamp) classes define project-specific payloads with `toJSON()` and `fromJSON()` serialization.
Multimodal alert method	Combine LED, buzzer, and LCD into a single coordinated guidance cue.	All three outputs are activated together in the "Reminder Active" state to reinforce the reminder across visual, auditory, and textual channels.
Hardware abstraction layer	Extend the object-oriented model to wrap each physical component as a class.	The hardware/firmware layer of the class diagram encapsulates each device, isolating control logic from the higher reminder and drawer-management logic.
Acrylic light-diffusion technique	Adapt the drawer material to act as part of the visual feedback system.	Transparent acrylic was selected so LED illumination spreads through the compartment, making the selected drawer easier to identify.
FSM modeled directly in LaTeX	Produce the firmware state diagram as reproducible, version-controlled source.	The state diagram is drawn with TikZ inside the design document rather than imported as an external image.

Table 5: Created and adapted tools and methods, with their function and evidence of use

4.1. Physical Prototype (Early Development)

The following figures show the laser-cut prototype of the Smart Pill Reminder Box during early development. They provide evidence of the application of the 3D modeling, laser cutting, and material selection tools described in previous sections.

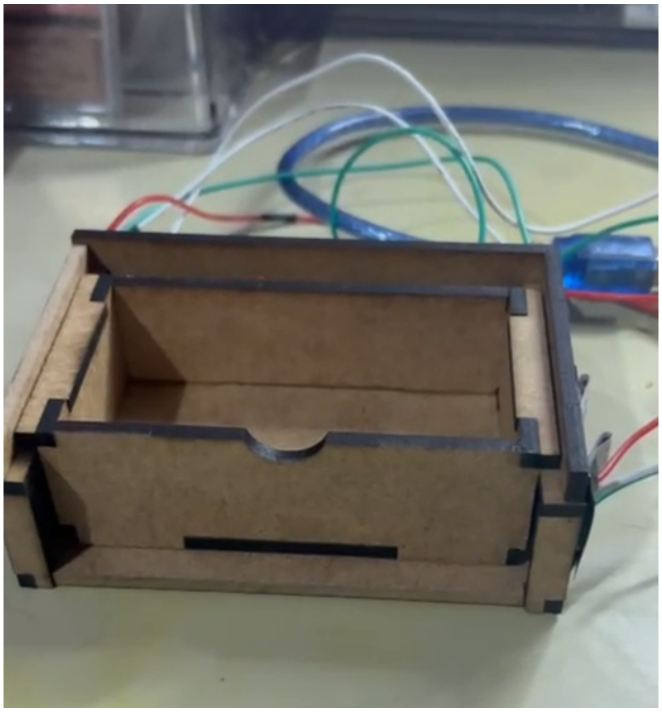

Figure 1: Laser-cut prototype with the reminder inactive (all outputs off)

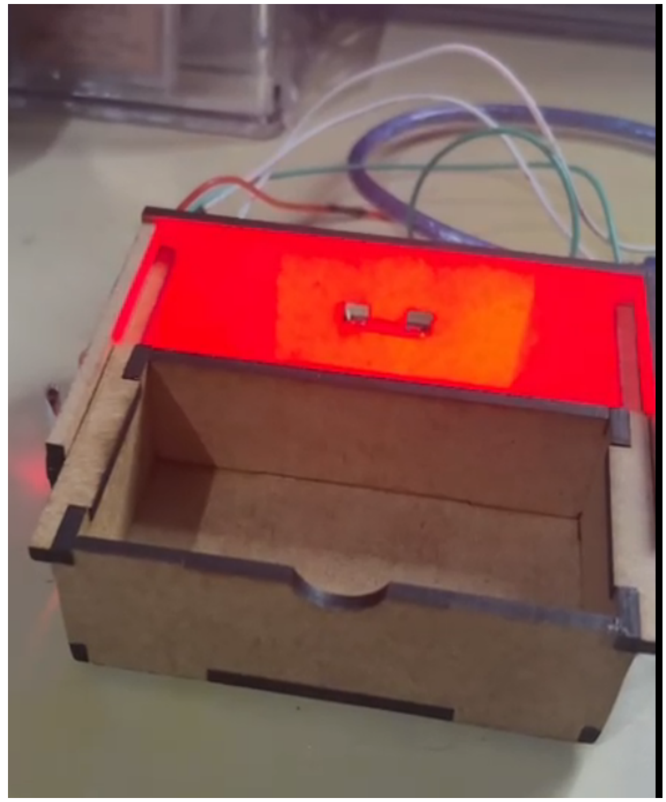

Figure 2: Laser-cut prototype with the reminder active (LED and buzzer on)